

My Home SOC Lab Project: Malware Simulation and Network Investigation

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My IT Certifications:

- Scrum Master
- Google Project Management & Agile
- IT Service Management

My SOC Certifications:

- ComptiaSec+
- Presecurity_TryHackMe
- Cyber Security 101_TryHackMe
- SOC Level 1 _TryHackMe
- Advent of Cyber 2025_ TryHackMe

1. Project Overview

This project simulates a **cyber attack and defensive investigation** in a **controlled home lab environment**.

An attacker machine **generates malware** using the **Metasploit Framework**, which is executed on a Windows victim machine. The attack is **monitored using** network and host-based security tools, including **Wireshark, Splunk, and Brim**.

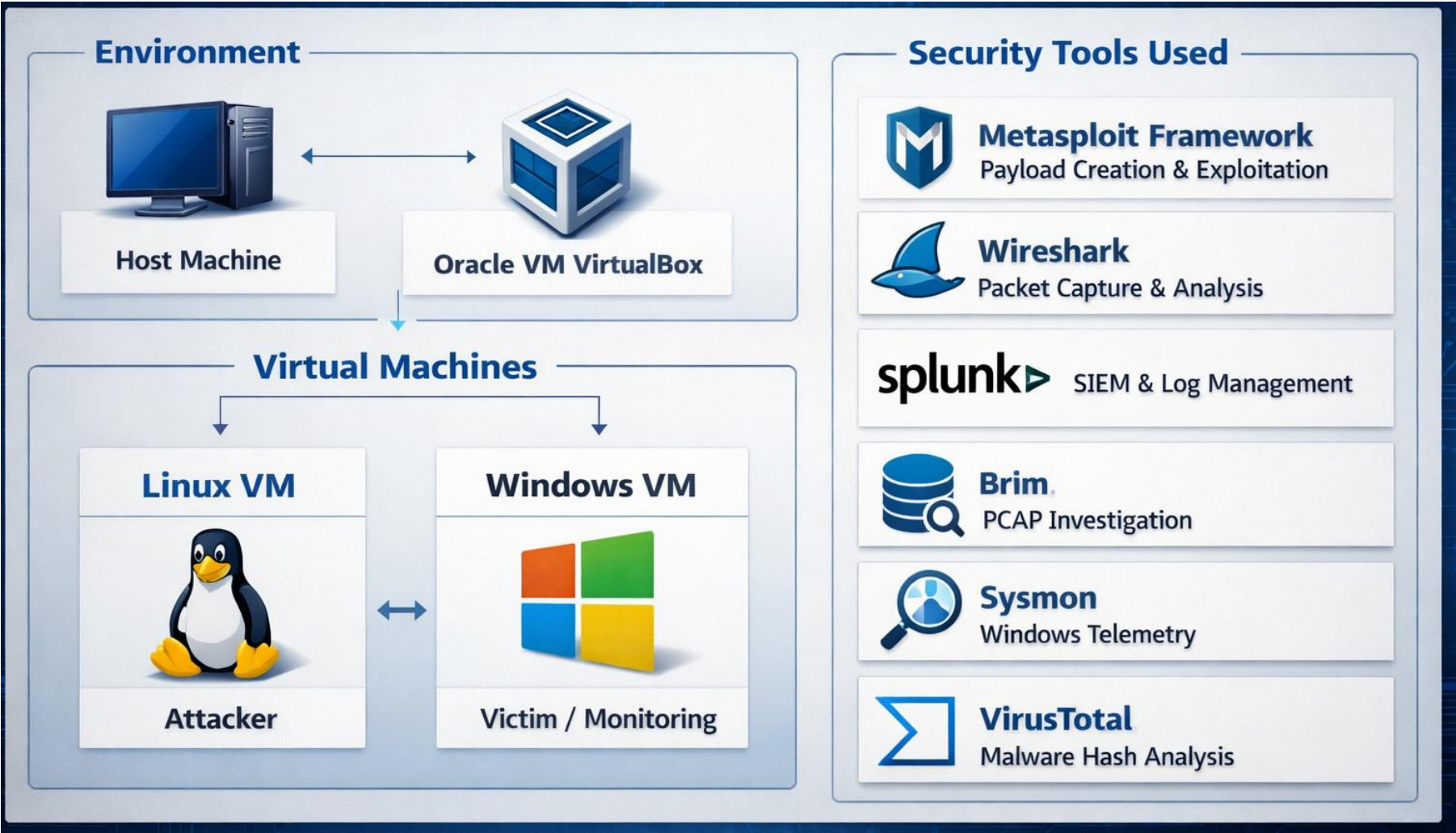
The **objective** was to:

- **Simulate a real attack** scenario
- Capture **network traffic** and system logs
- **Investigate indicators** of compromise
- **Practice SOC** analyst workflows
- **Analyze large packet** capture datasets



The lab was implemented using Oracle VM VirtualBox virtualization

2. Lab Architecture



3. Attack Simulation

A **malicious payload** was generated using the **Metasploit** Framework and **delivered** to the **Windows victim** machine.

Steps

- **Generate** reverse shell payload
- **Deliver** executable to Windows VM
- **Execute** payload on victim machine
- Establish **connection** back to attacker machine
- **Monitor** network and system activity

This created observable security events for analysis.



4. Detection and Monitoring

Multiple security **monitoring tools** were used to **observe attack behavior**.

Network Monitoring

Using Wireshark to capture:

- TCP sessions
- suspicious outbound connections
- payload communication

Large .pcap files were generated during monitoring.

Host Monitoring

The Windows machine generated logs through:

- Sysmon process creation
- network connections
- file activity

These logs were ingested into Splunk for analysis.



5. Investigation of Network Traffic

Due to the **size of packet capture files**, manual inspection using Wireshark became inefficient. To address this, PCAP **data was analyzed using**:

Brim

Brim allowed efficient investigation through:

- indexed queries
- Zeek logs
- network metadata extraction

Example Investigation Questions

- Which IP initiated the connection?
- What protocol was used for the reverse shell?
- Were suspicious DNS queries generated?
- What ports were used for command-and-control communication?



6. Incident Response Simulation

During the project, a **misconfiguration was discovered** where the **virtual machines were not fully isolated from external networking**.

Actions taken:

- Restored VM snapshots
- Performed host system log analysis
- Monitored network activity for suspicious traffic
- Verified system integrity using logs and telemetry



This simulated a real-world **incident response and containment process**.

7. Skills Demonstrated

This project demonstrates practical skills in:

Security Operations

- Security monitoring
- SIEM analysis
- Log correlation

Network Security

- Packet capture analysis
- TCP session investigation
- Threat detection

Incident Response

- Threat containment
- Log analysis
- Environment remediation

Security Tooling

Experience using:

- Splunk
- Wireshark
- Brim
- Metasploit Framework



8. Additional Practice

To strengthen investigation skills, several security labs were completed on TryHackMe including:

- Brim Investigation Lab
- Masterminds
- Warzone

These exercises focused on:

- Malware traffic analysis
- Network threat hunting
- Forensic investigation of packet captures



9. Lessons Learned

Key lessons from this project:

- **Proper network isolation is critical (Sandboxing)** when testing malware.
- **Large packet** capture files **require** specialized **analysis tools**.
- **Combining** host **telemetry and network** logs improves detection.
- **SIEM** platforms are essential **for correlating security events**.

10. Future Improvements

Potential enhancements to this lab include:

- Integrating **Zeek** for automated network logging
- Adding additional **endpoints** for multi-host detection
- **Automating log ingestion** pipelines
- **Simulating advanced attacker** techniques

11. Summary and References

Summary

Home SOC Lab – Malware Simulation & Network Investigation

- Built a cybersecurity lab using Oracle **VM VirtualBox** with attacker and victim machines
- Generated and executed malware using **Metasploit** Framework
- Monitored attack activity using **Wireshark** and **Splunk**
- Investigated large network captures using **Brim**
- Correlated host telemetry (**Sysmon**) and network logs to detect indicators of compromise

References

https://github.com/hyphennR/Document_SOC_Home_Lab

<https://github.com/xAHIINX00/SOC-Home-Lab-Attack-Defense-Simulation>

<https://github.com/praiseordu/SOC-ANALYSIS-LAB>

<https://diog0net0sec.medium.com/how-i-built-my-virtual-soc-analyst-home-lab-9b1e6baa5c53>

<https://www.youtube.com/watch?v=kku0fVfksrk&list=PLG6KGSNK4PuBWmX9NyKU0wnWamjxdKhDJ&index=1&pp=iAQB>

**Thank you & good luck
with your SOC Lab Project**